

NAME SCHOOL TEACHER 

Pre-Junior Certificate Examination, 2015

Mathematics

Foundation Level

Time: 2 hours

300 marks

School stamp

Running total	
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For examiner			
Question	Mark	Question	Mark
1		11	
2		12	
3		13	
4		14	
5		15	
6		16	
7			
8			
9			
10		Total	

Grade

Instructions

There are 16 questions on this examination paper. Answer **all** questions.

Questions do not necessarily carry equal marks. To help you manage your time during this examination, a maximum time for each question is suggested. If you remain within these times, you should have about 10 minutes left to review your work.

Write your answers in the spaces provided in this booklet. You may lose marks if you do not do so. There is space for extra work at the back of the booklet. You may also ask the superintendent for more paper. Label any extra work clearly with the question number and part.

The superintendent will give you a copy of the *Formulae and Tables* booklet. You must return it at the end of the examination. You are not allowed to bring your own copy into the examination.

You will lose marks if all necessary work is not clearly shown.

You may lose marks if the appropriate units of measurement are not included, where relevant.

You may lose marks if your answers are not given in simplest form, where relevant.

Write the make and model of your calculator(s) here:

Question 1

(Suggested maximum time: 5 minutes)

(a) Find the value of:

(i) $24 - 16 =$

(ii) $167.3 + 47.2 =$

(iii) $45.9 \times 2.3 =$

(iv) $76.2 \div 0.4 =$

[illegible]

(b) The numbers in the table below are shown in fraction, decimal and percentage form. Complete the table.

Fraction	Decimal	Percentage
$\frac{1}{2}$	0.5	
	0.9	90%
$\frac{3}{4}$		75%

[illegible]

(c) List all the factors of 18.

[illegible]

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Question 2 (Suggested maximum time: 5 minutes)

Question 2 (Suggested maximum time: 5 minutes)

Roy has four cards. Each card has a number written on it as shown:

The cards can be rearranged to make different four-digit numbers, for example:

Diagram showing four boxes containing the numbers 3, 6, 1, and 7.

- (a)** Use all four cards to show:

- (i) the smallest possible number

- (ii) the largest possible **even** number.

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- (b) (i)** Roy uses all four cards to make a true statement.
Write the number on the cards to make this true. Use each card only once.

$$\boxed{} + \boxed{} = \boxed{ }$$

[illegible]

- (ii)** Use all four cards to show the number with the closest value to 2000.

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- (c) A fifth card is needed to show the result of the multiplication 5821×3 . Write the number that should be on the fifth card.

Fifth Card =

7

[illegible]

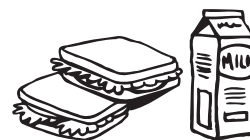
Question 3

(Suggested maximum time: 10 minutes)

A school canteen has the following menu:

Food	
Soup	€1.45
Sandwich Roll	€2.25
Toasted Sandwich	€2.45

Drinks	
Tea	€1.20
Coffee	€1.35
Milk	€0.90
Juice	€1.40



- (a)** Ann buys a sandwich roll and a juice.
How much does she pay?

[illegible]

- (b)** Sean buys a soup, two toasted sandwiches and a tea.
How much change does he receive from €10?

[illegible]

- (c) Laura has €3.40. She wishes to buy one item of food and one drink. List **one** combination she could buy.

[illegible]

- (d)** The canteen has a ‘Special Offer’ promotion as follows:

Special Offer
Soup, Toasted Sandwich and Juice. All for €5.

Gerard chooses the 'Special Offer'.
How much does he save?

[illegible]

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Question 4

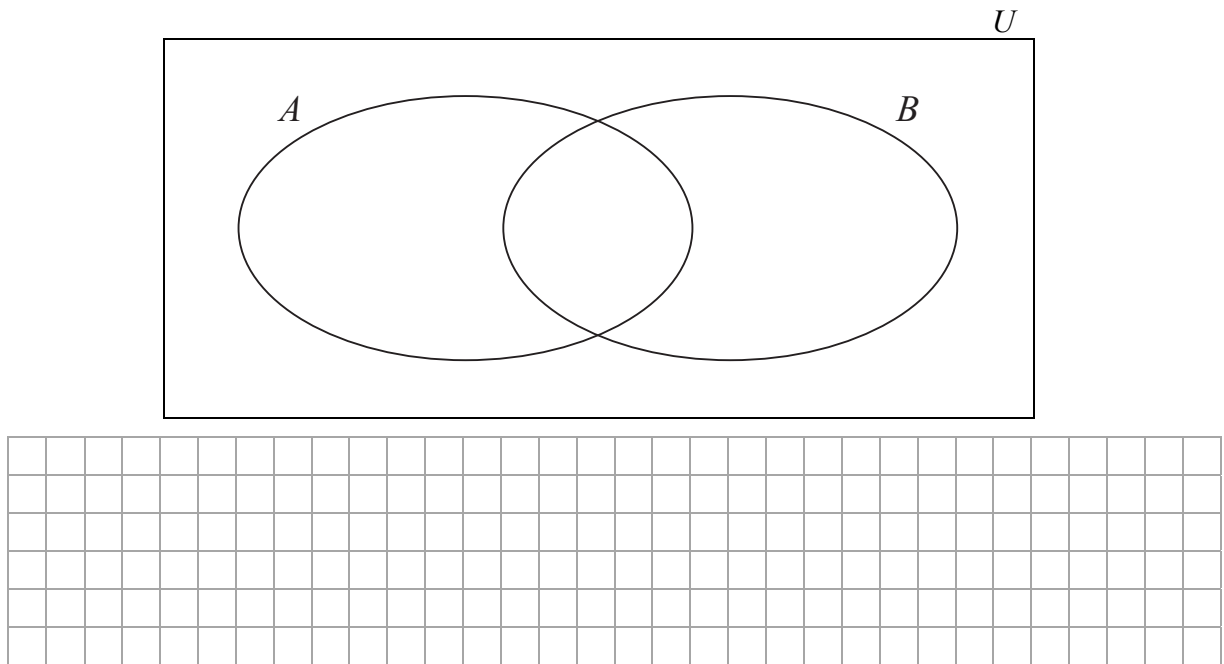
(Suggested maximum time: 5 minutes)

$$U = \{1, 2, 3, 4, 5, 6, 7\}.$$

$$A = \{1, 3, 5, 7\}.$$

$$B = \{2, 3, 4, 7\}.$$

(a) Fill in the Venn diagram below.



(b) (i) List the elements of $A \cup B$.

$$A \cup B = \{ \quad , \quad , \quad , \quad , \quad , \quad \}$$

(ii) List the elements of $A \cap B$.

$$A \cap B = \{ \quad , \quad \}$$

(c) Find $\#(A \cap B)$.

Question 5

(Suggested maximum time: 5 minutes)

Aidan wishes to buy an Easter egg.

He has three different sizes of eggs to choose from.

He has three different types of chocolate to choose from.

These are shown in the tables below.



Size	Type
Small	Milk
Medium	Dark
Large	White

- (a) Complete the table below to show all of the nine possible Easter eggs that Aidan can choose.

	Size	Type
1	Small	Milk
2		
3		
4		
5		
6		
7		
8		
9		

- (b)** If Aidan selects an Easter egg at random, what is the probability that he chooses a large milk chocolate Easter egg?

[illegible]

Question 6

(Suggested maximum time: 15 minutes)

Students in a class were asked how many text messages they sent before school.
The results are as follows.



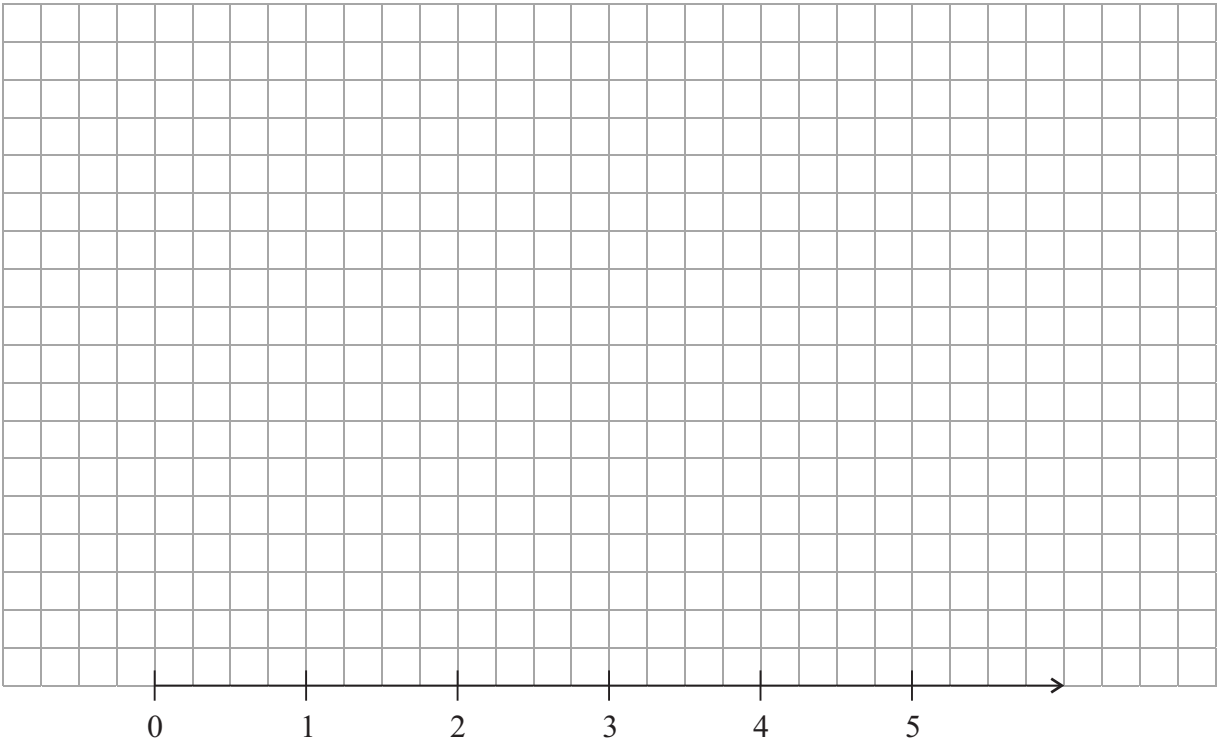
3 1 4 0 3 2 3 2 1
2 3 1 3 0 1 3 1 3

(a) Complete the table below to show the results.

Number of text messages sent	Tally	Frequency
0		
1		
2		
3		
4		
Total		

(b) How many students are in the class?

(c) Represent the data on a line plot.



- (d) Write down the **mode** of the data.
Give a reason for your choice.

[illegible]

- (e) Tom says that the **mean** number of text messages sent by the students was 2. Show that he is correct.

[illegible]

- (f) The cost to the students of sending all the text messages was €1.26, plus VAT at 23%. Find the total cost of sending all the text messages.

Cost of text messages	€
VAT @ 23%	€
Total Cost	€

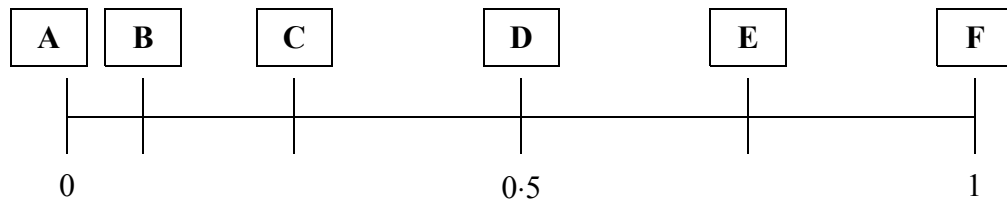
[illegible]

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Question 7 (Suggested maximum time: 5 minutes)

Question 7 (Suggested maximum time: 5 minutes)

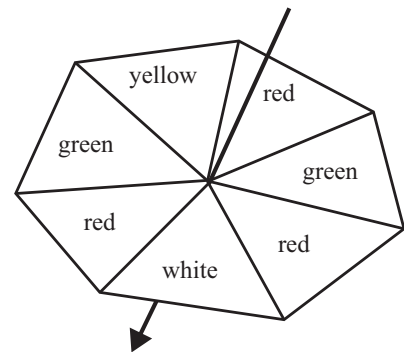
- (a)** The probability that each of the events A, B, C, D, E and F will happen is shown on the probability scale below.



Match the event with the term which best describes its probability.
The first one has been done for you.

Term	Event
Likely	E
Even chance	
Almost impossible	
Certain	
Unlikely	
Impossible	

- (b)** A fair spinner has seven equal sectors.
Each of the sectors is white, red, green or yellow.



To play the game, a player selects a colour and spins the arrow once. The arrow must stop on the colour the person has selected to win.

- (i) On which colour is the spinner most likely to land?
Give a reason for your answer.

[illegible]

- (ii)** Write down the probability that the spinner will land on green.

Question 8 (Suggested maximum time: 5 minutes)

Question 8 **(Suggested maximum time: 5 minutes)**

Dan cycled from his home to a shop. Afterwards he returned home by the same route.

The graph below shows Dan's complete journey to and from the shop.



- _____

- | |
|--|
| |
|--|

- [illegible]

- [illegible]

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Question 9

(Suggested maximum time: 5 minutes)

- (a)** The first four numbers of a pattern are

2, 8, 14, 20

- (i)** Write down the next two terms in the pattern.

Answer = ,

- (ii)** Write down, in words, a rule for continuing the pattern.

[illegible]

- (b)** Select the expression that matches each of the following:

- (i) three more than x

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$3x$	$x+3$	$x-3$	x^3
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- (ii) four less than x

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$\frac{x}{4}$	$4 - x$	$x - 4$	$4x$
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- (iii) two times x

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x^2	$x + 2$	$\frac{x}{2}$	$2x$
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- (iv) half of x

$\frac{2}{x}$	$x - 2$	$2x$	$\frac{x}{2}$
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Question 10

(Suggested maximum time: 5 minutes)

- (a)** Find the value of $5x - 2y$, where $x = 3$ and $y = 2$.

[illegible]

- (b)** Simplify $6x + 4y + 7x - 3y$.

[illegible]

- (c) Solve $4(x - 3) = 20$.

[illegible]

- (d) Factorise $15x - 21$.

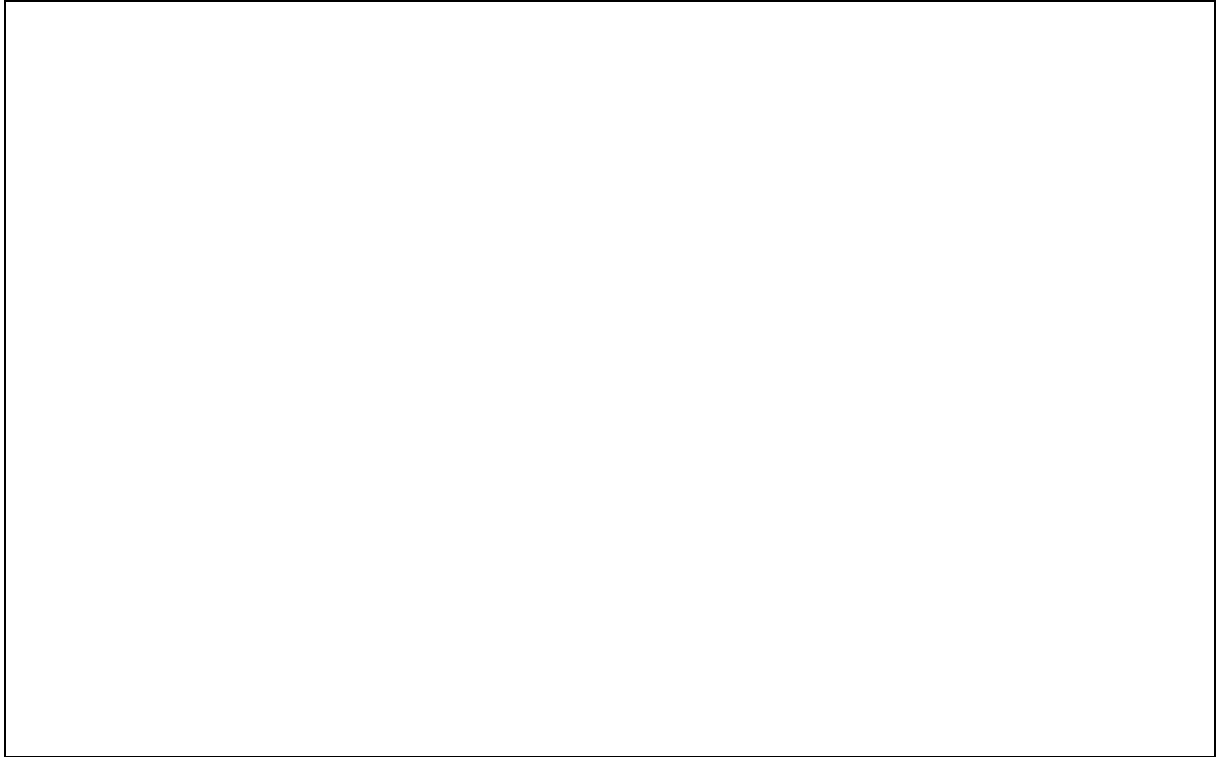
$$-15x - 21 = 3(\quad) -$$

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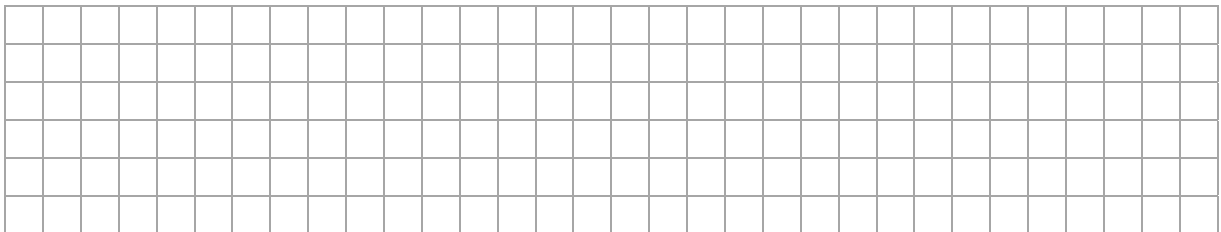
Question 11

(Suggested maximum time: 15 minutes)

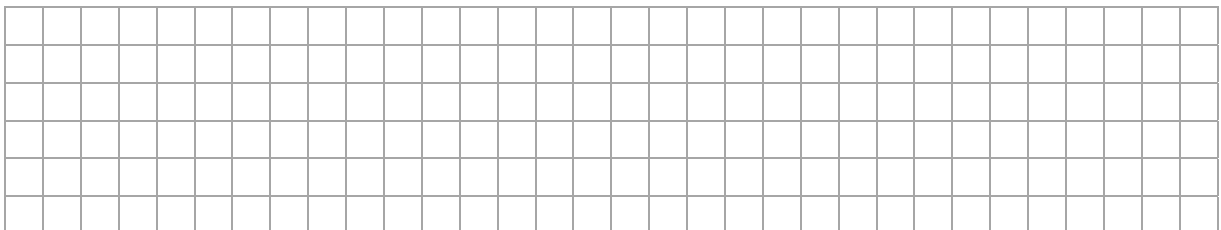
- (a)** Construct a rectangle 8 cm long and 6 cm wide.



- (b)** Find the **perimeter** of the rectangle in cm.



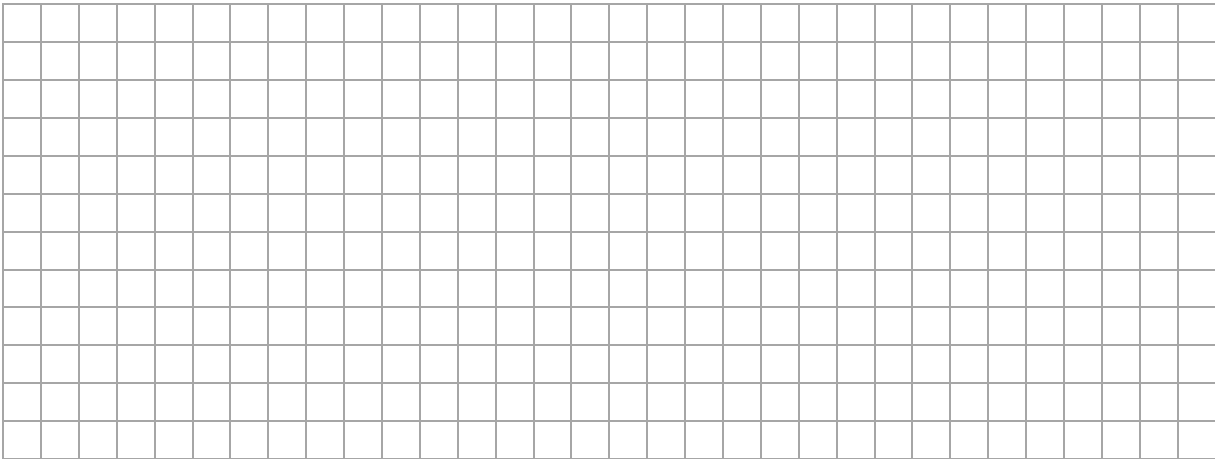
- (c)** Find the **area** of the rectangle in cm^2 .



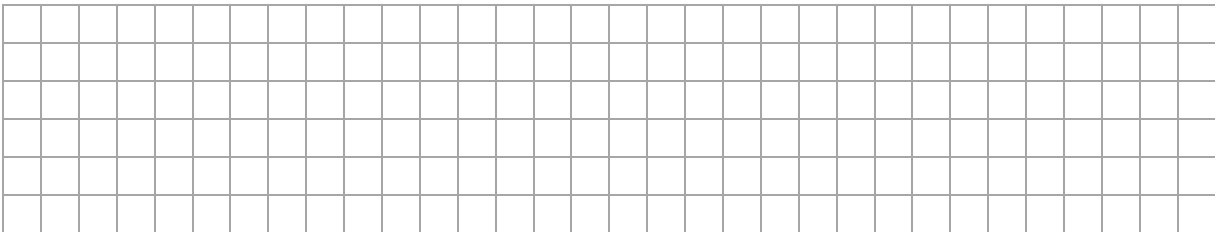
- (d)** **(i)** Draw in a diagonal of the rectangle you have constructed in part **(a)**.
- (ii)** Measure and write down the length of the diagonal you have drawn in part **(d)(i)**.
Give your answer in centimetres.

Length of diagonal =

(e) Use the Theorem of Pythagoras to show that you measured the length of the diagonal correctly.



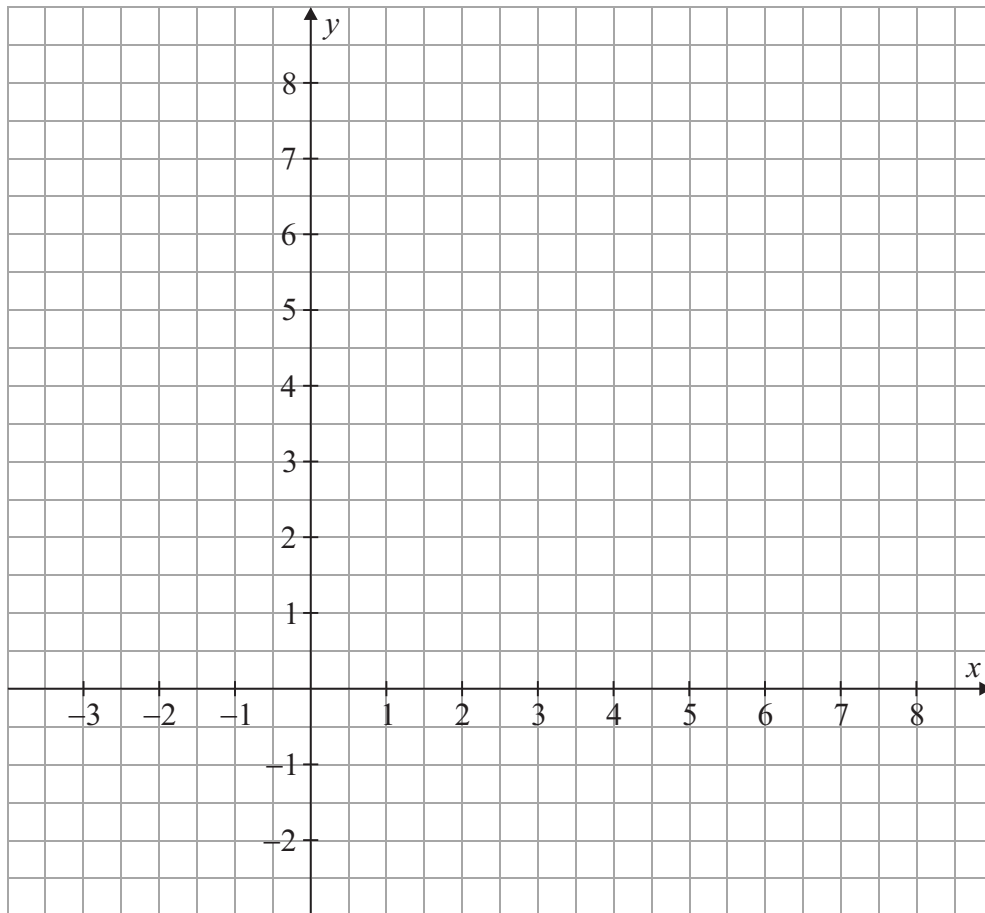
(f) Write down **one** property of a rectangle.



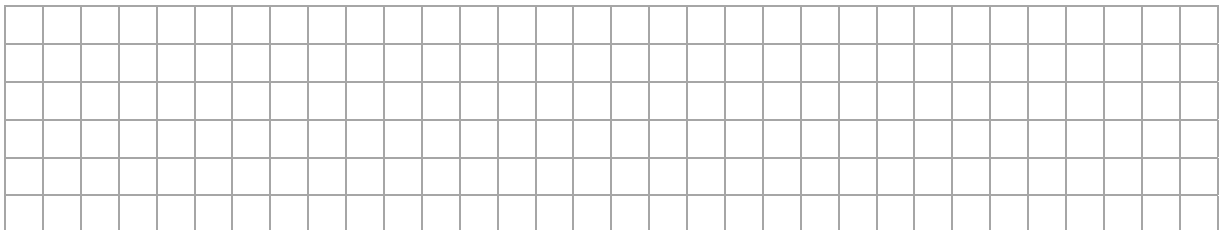
Question 12**(Suggested maximum time: 10 minutes)**

$A(-1, 1)$, $B(5, 3)$ and $C(0, 8)$ are three points.

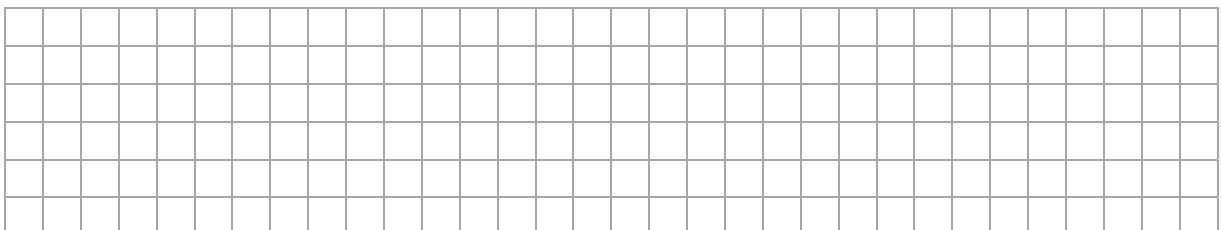
- (a) Plot the points A , B and C on the co-ordinated plane below.



- (b) Join the three points to form a triangle.
- (c) On your diagram above, draw the axis of symmetry of the triangle.
- (d) Find the midpoint of $[AB]$.



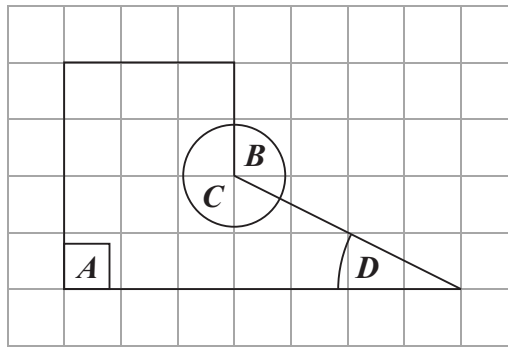
- (e) Find the slope of AB .



Question 13

(Suggested maximum time: 5 minutes)

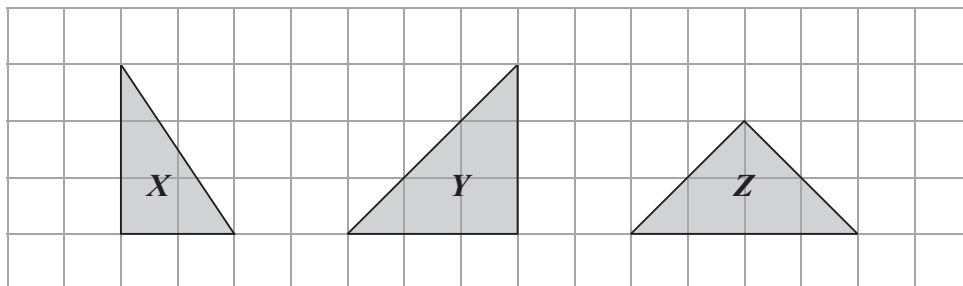
- (a)** The diagram below shows different types of angles.



Write down the letter which represents each of the following angles in the diagram.

Angle	Letter
Right angle	
Obtuse angle	
Acute angle	
Reflex angle	

- (b)** Three triangles, X , Y and Z , are shown on the grid below. Each square of the grid is 1 cm in length.



- (i) Which two triangles are isosceles triangles?
Give a reason for your answer.

and

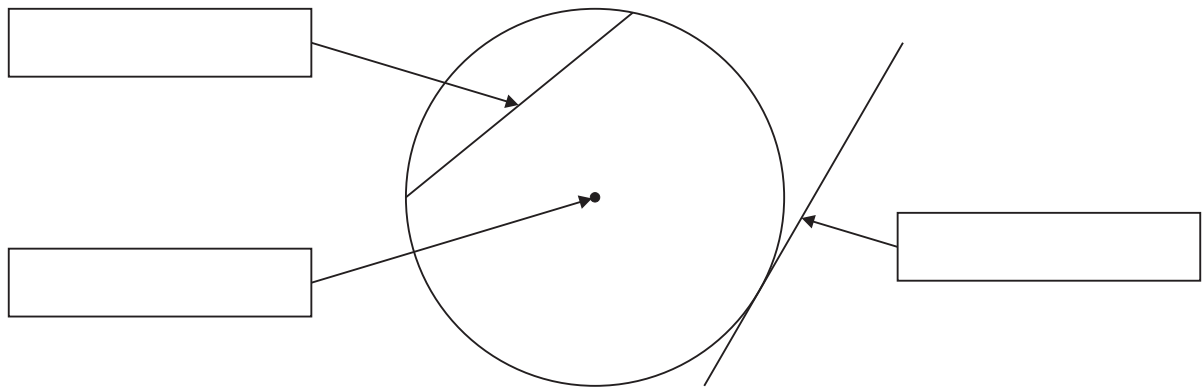
Reason:

- (ii)** Find the area of the triangle with the greatest area in cm^2 .

[illegible]

(Suggested maximum time: 5 minutes)

(a) The diagram below shows different elements of a circle.



On the diagram, fill in the correct term for each of the elements indicated from the following table.

Centre	Diameter	Chord	Tangent
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- (b)** The radius of a circle is 14 cm.
Find the circumference of the circle. Use $\pi = 3.142$.

[illegible]

(Suggested maximum time: 5 minutes)

- (a)** f is the function $f: x \mapsto 7x - 3$, where $x \in \mathbb{R}$. Find $f(2)$.

[illegible]

- (b)** $\{(1, 2), (2, 4), (3, 6), (4, 8)\}$ are four couples of a function g .

- (i) Write down the domain and range of the function g .

$$\text{Domain} = \left\{ \begin{array}{c} \text{ } \end{array} \right\}$$
$$\text{Range} = \left\{ \begin{array}{c} \text{ } \end{array} \right\}$$

- (ii) Explain the relationship between the domain and the range of the function g .

[illegible]

Question 16

(Suggested maximum time: 5 minutes)

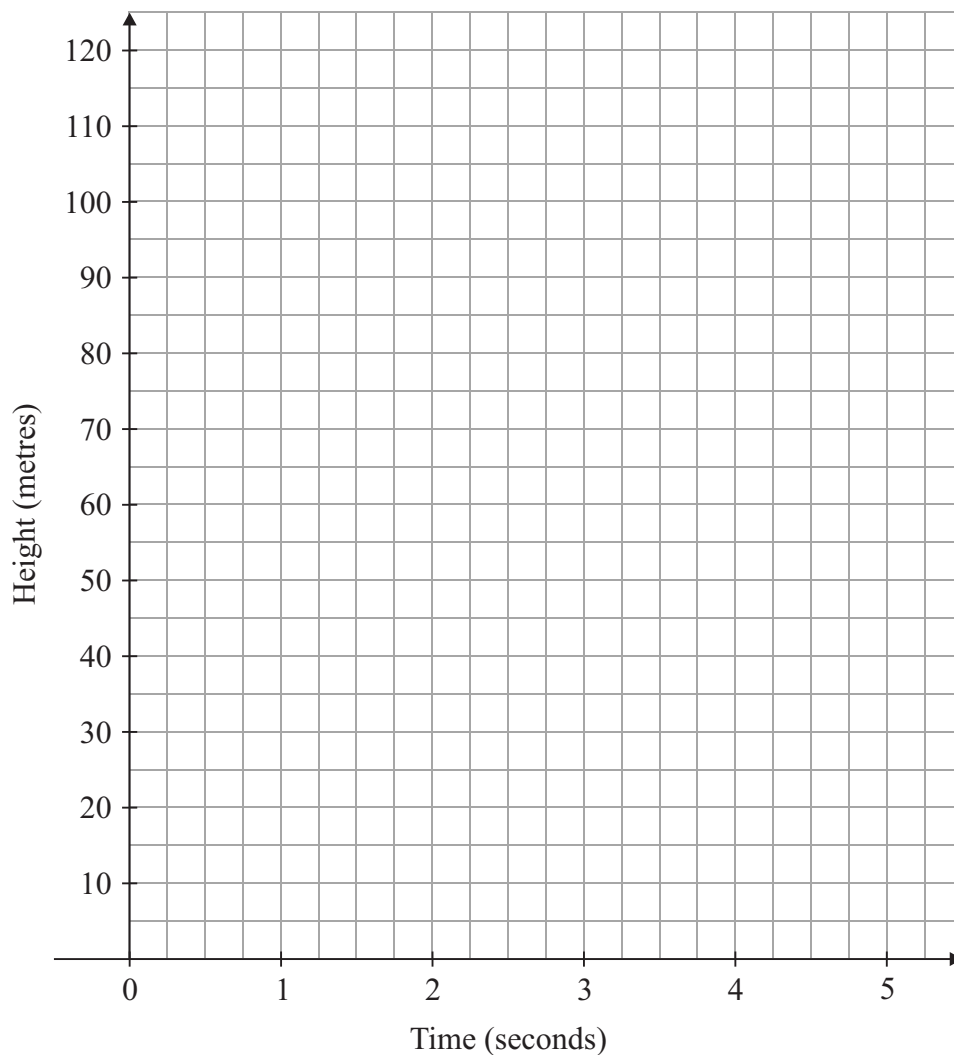
A firework is launched straight up into the air.

The table below shows the height of the firework, to the nearest metre, over the first 4 seconds.

Time (seconds)	Height (metres)
0	0
1	50
2	85
3	105
4	110



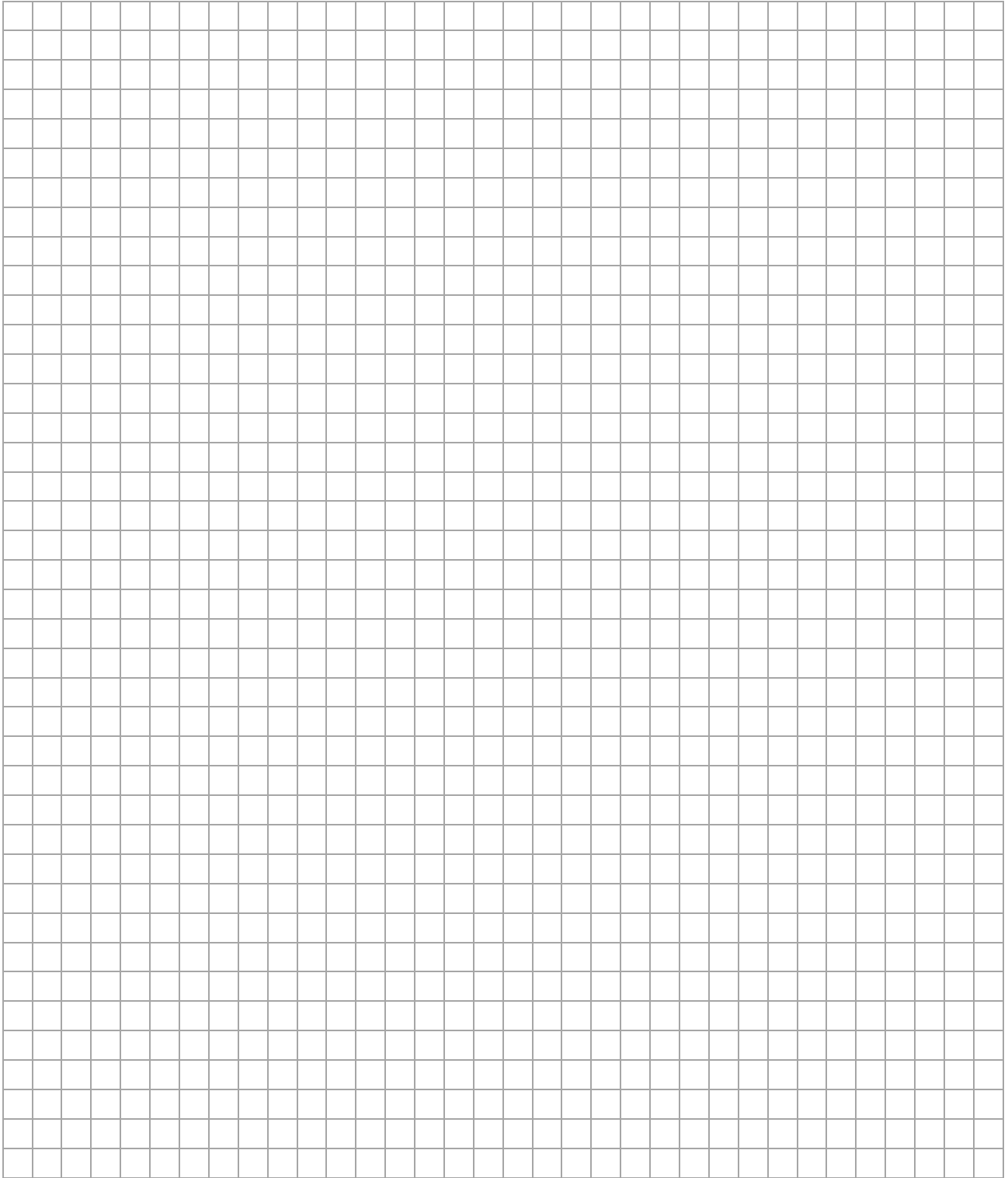
- (a) Draw a graph to show the height of the firework over the first 4 seconds.



- (b) After exactly 4 seconds, the firework rocket is at its highest point.
At what height do you expect the firework rocket to be after 5 seconds?

Height after 5 seconds =

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Pre-Junior Certificate, 2015 – Foundation Level

Mathematics

Time: 2 hours

